Comptia A+ Assignment

[A+ - Troubleshooting and Helpdesk]

Diagnosing Hardware Failures.

Common hardware issues and their symptoms. Test components using a multimeter.

Task

- **1. List 5 common hardware issues you might encounter in a desktop PC and write the typical symptom for each (for example: 'No display on monitor' — symptom: blank screen, no POST beep).**

  - Hardware failures are common in desktop computers and can prevent the system from starting or working properly. The table below lists five common hardware issues along with their typical symptoms.

Hardware Issue	Typical Symptom
No Display on Monitor	The monitor remains blank, and the computer may not complete the POST process or display any video output.
Faulty RAM (Memory)	The computer may produce continuous or repeated beep codes, restart automatically, freeze, or fail to boot.

Hard Disk / SSD Failure	The operating system does not load, the BIOS may not detect the storage device, or error messages such as "No Boot Device Found" appear.
Power Supply (SMPS) Failure	The computer does not power on, fans do not spin, and no indicator lights appear on the motherboard or cabinet.
Overheating CPU or CPU Fan Failure	The computer suddenly shuts down, restarts unexpectedly, runs very slowly, or displays a CPU Fan Error warning during startup.

❖ **2. Given this scenario: A laptop is not charging even when plugged in. Write down a step-by-step troubleshooting plan using only basic tools (no software diagnostics), and explain what you would check at each step.**

➢ A laptop is not charging even though the charger is connected given the below Scenario for Step-by-Step Troubleshooting Plan.

➢ Step 1: Check the Power Outlet:
- Plug another device (such as a phone charger or lamp) into the same power outlet to make sure the socket is supplying electricity.

➢ What to Check:
- The power outlet is working properly.
- The power switch is turned ON.

- Step 2: Inspect the Charger and Power Cable:
  - Examine the laptop charger and cable for any cuts, burns, loose connections, or physical damage.
- What to Check:
  - Adapter LED is ON (if available).
  - Cable is not broken or damaged.
  - Charger connector is not bent.

- Step 3: Check the Laptop Charging Port:
  - Inspect the charging port for dust, dirt, or bent pins.
- What to Check:
  - Charging port is clean.
  - Connector fits securely.
  - No physical damage is visible.

- Step 4: Check the Battery Connection:
  - If the laptop has a removable battery, remove it and reinstall it properly.
- What to Check:
  - Battery is correctly seated.
  - Battery contacts are clean.
  - Battery is not swollen or damaged.

- Step 5: Test with Another Compatible Charger:
  - Use another charger with the correct voltage and connector to verify whether the original charger is faulty.
- What to Check:

- Laptop starts charging with the replacement charger.
- Original charger may need replacement.

➢ Step 6: Test Without the Battery (If Removable):
- Remove the battery and connect only the charger to the laptop.

➢ What to Check:
- If the laptop powers on, the battery may be faulty.
- If it still does not power on, the charger or motherboard power circuit may be the problem.

➢ Step 7: Check the Charging Indicator:
- Observe the charging LED on the laptop after connecting the charger.

➢ What to Check:
- Charging light turns ON.
- Battery charging indicator appears.
- Any blinking light pattern that may indicate a hardware fault.

➢ After checking the power outlet, charger, charging port, battery, and charging indicator, the faulty component can usually be identified. Replacing the damaged charger or battery often resolves the charging problem.

❖ **3. Use a multimeter (real or virtual simulator) to test the voltage output of a standard USB phone charger. Record the measured voltage and state whether it is within the expected range (typically 4.75V to 5.25V for USB).**

➢ To measure the output voltage of a standard USB phone charger using a multimeter and verify whether it is within the standard USB voltage range.

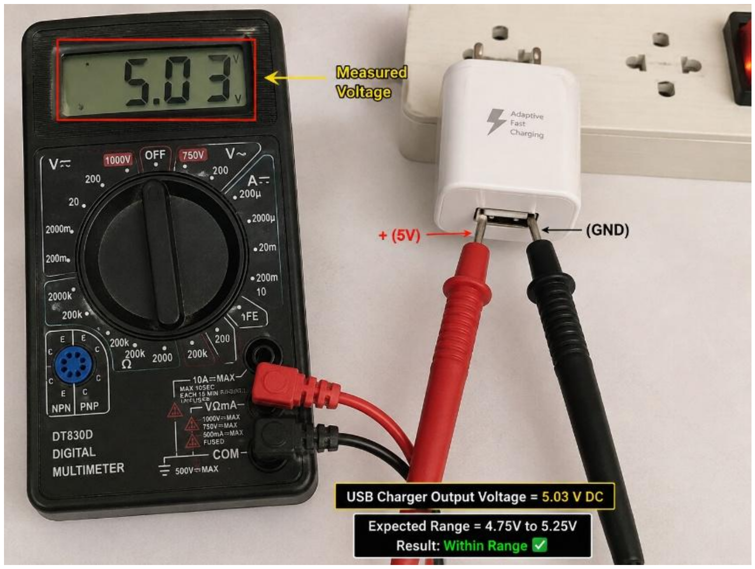
➢ Tools Required:
- Digital Multimeter (or Virtual Multimeter Simulator)
- Standard USB Phone Charger (5V)
- USB Cable / USB Test Adapter (if available)

➢ Step-by-Step Procedure:

➢ Step 1: Prepare the Equipment:
- Connect the USB phone charger to a power outlet and switch it ON.

➢ Step 2: Set the Multimeter:
- Set the digital multimeter to DC Voltage (V⎓) mode.

➢ Step 3: Measure the USB Output:
- Place the red probe on the +5V (VBUS) pin and the black probe on the Ground (GND) pin of the USB output.



- **4. A PC is randomly shutting down. Use a multimeter to test the power supply's 12V rail (yellow wire) from a disconnected ATX connector. Describe the steps and write down the voltage you find. If the voltage is below 11.5V, what hardware issue could this indicate? Remember to set the multimeter to DC voltage and probe between the yellow and black wires.**

  - To measure the 12V output of a desktop PC SMPS using a digital multimeter and determine whether the power supply is functioning correctly.
  - <u>Tools Required:</u>
    - <u>Digital Multimeter</u>
    - <u>Desktop PC SMPS (ATX Power Supply)</u>
    - <u>24-pin ATX Connector</u>

- <u>Step-by-Step Procedure:</u>

- <u>Step 1: Turn Off the Power:</u>
  - Switch off the computer and disconnect the SMPS from the motherboard.

- <u>Step 2: Set the Multimeter:</u>
  - Set the digital multimeter to <u>DC Voltage (V⎓)</u> mode.

- <u>Step 3: Identify the Wires:</u>
  - Yellow Wire = +12V
  - Black Wire = Ground (GND)

- <u>Step 4: Measure the Voltage:</u>
  - Touch the <u>red probe</u> to the <u>yellow (+12V)</u> wire and the <u>black probe</u> to the <u>black (GND)</u> wire on the disconnected ATX connector.